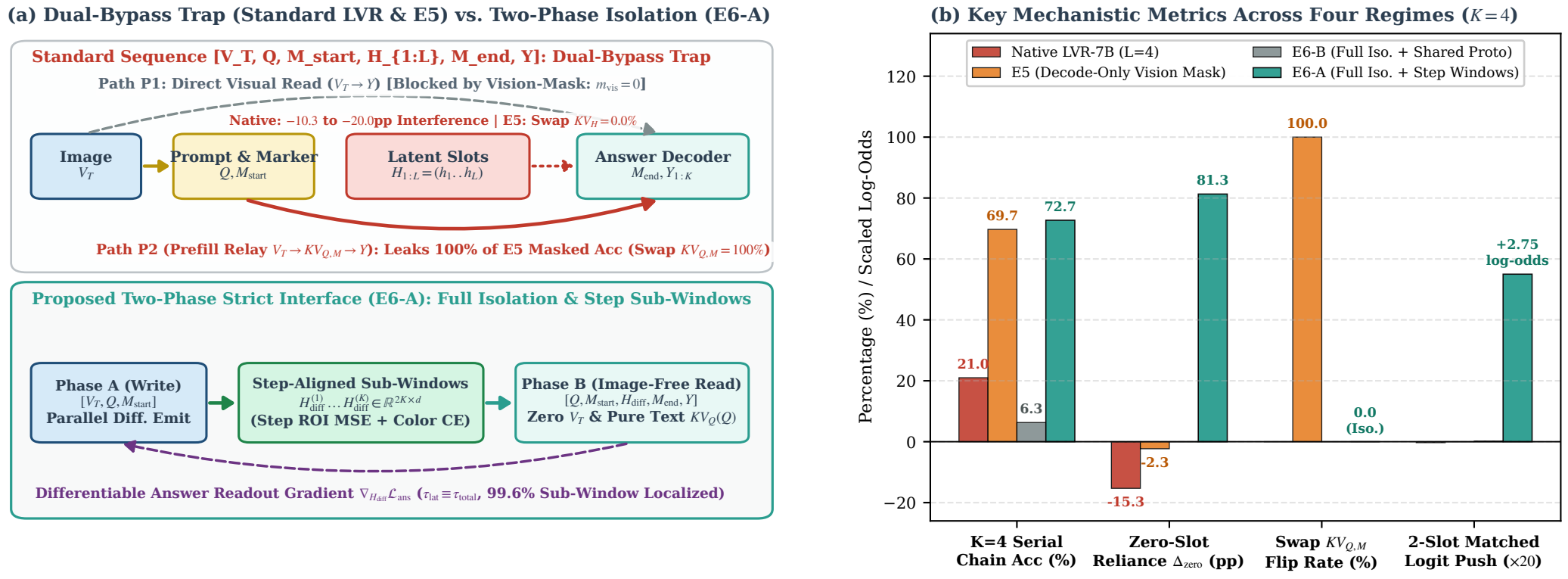
(a) Dual-Bypass Trap (Standard LVR & E5) vs. Two-Phase Isolation (E6-A)
Standard Sequence [V_T, Q, M_start, H_{1:L}, M_end, Y]: Dual-Bypass Trap
Path P1: Direct Visual Read ($V_T \to Y$) [Blocked by Vision-Mask: $m_{vis}=0$]
Native: −10.3 to −20.0pp Interference | E5: Swap $KV_H = 0.0\%$
Image
V_T
Prompt & Marker
Q, M_{start}
Latent Slots
$H_{1:L} = (h_1 .. h_L)$
Answer Decoder
$M_{end}, Y_{1:K}$
Path P2 (Prefill Relay $V_T \to KV_{Q,M} \to Y$): Leaks 100% of E5 Masked Acc (Swap $KV_{Q,M} = 100\%$)
Proposed Two-Phase Strict Interface (E6-A): Full Isolation & Step Sub-Windows
Phase A (Write)
$[V_T, Q, M_{start}]$
Parallel Diff. Emit
Step-Aligned Sub-Windows
$H^{(1)}_{diff} \dots H^{(K)}_{diff} \in \mathbb{R}^{2K \times d}$
(Step ROI MSE + Color CE)
Phase B (Image-Free Read)
$[Q, M_{start}, H_{diff}, M_{end}, Y]$
Zero V_T & Pure Text $KV_Q(Q)$
Differentiable Answer Readout Gradient $\nabla_{H_{diff}} \mathcal{L}_{ans}$ ($\tau_{lat} \equiv \tau_{total}$, 99.6% Sub-Window Localized)
(b) Key Mechanistic Metrics Across Four Regimes ($K=4$)
Native LVR-7B (L=4)
E5 (Decode-Only Vision Mask)
E6-B (Full Iso. + Shared Proto)
E6-A (Full Iso. + Step Windows)
Percentage (%) / Scaled Log-Odds
120
100
80
60
40
20
0
−20
21.0
69.7
6.3
72.7
-15.3
-2.3
81.3
100.0
0.0
(Iso.)
+2.75
log-odds
K=4 Serial
Chain Acc (%)
Zero-Slot
Reliance Δ_{zero} (pp)
Swap $KV_{Q,M}$
Flip Rate (%)
2-Slot Matched
Logit Push (×20)